Demo Study B: Gradient Boosting for Multi-site Rain Attenuation Estimation

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SYNTHETIC DEMO DOCUMENT - fictional content for software testing. Not a real publication.

Abstract

We investigate gradient boosting with XGBoost for rain attenuation estimation across five ground stations in tropical and temperate climates. Cross-site validation shows moderate performance loss on unseen locations.

Introduction

Satellite link planning requires attenuation statistics for many locations. This work examines whether a single machine learning model can serve multiple sites without retraining.

Data

The dataset combines beacon measurements from five stations, covering 3 years in total. Reanalysis humidity and temperature were added as features. Rain gauge data were available at three of the five sites.

Methodology

An XGBoost regressor is trained using leave-one-site-out cross-validation to test transferability across unseen sites. Hyperparameters were tuned by Bayesian optimization. We compare against a random forest and the ITU-R P.618 model.

Results

XGBoost reaches an RMSE of 1.05 dB on seen sites and 1.61 dB on held-out sites, improving over the ITU-R P.618 model, which obtains an RMSE of 2.10 dB on held-out sites. MAE is reduced by 31 percent relative to the random forest.

Limitations

Extreme events above 20 dB are underrepresented, and the model degrades under heavy rain. Uncertainty estimates were not produced; only point predictions are available.